Lab - Exploring DNS Traffic (Instructor Version)

Instructor Note: Red font color or gray highlights indicate text that appears in the instructor copy only.

Objectives

Part 1: Capture DNS Traffic

Part 2: Explore DNS Query Traffic

Part 3: Explore DNS Response Traffic

Background / Scenario

Wireshark is an open source packet capture and analysis tool. Wireshark gives a detailed breakdown of the network protocol stack. Wireshark allows you to filter traffic for network troubleshooting, investigate security issues, and analyze network protocols. Because Wireshark allows you to view the packet details, it can be used as a reconnaissance tool for an attacker.

In this lab, you will install Wireshark and use Wireshark to filter for DNS packets and view the details of both DNS query and response packets.

Required Resources

- 1 PC with internet access and Wireshark installed

Instructor Note: Using a packet sniffer such as Wireshark may be considered a breach of the security policy of the school. It is recommended that permission is obtained before running Wireshark for this lab. If using a packet sniffer such as Wireshark is an issue, the instructor may wish to assign the lab as homework or perform a walk-through demonstration.

Instructions

Part 1: Capture DNS Traffic

Step 1: Download and install Wireshark.

a. Download the latest stable version of Wireshark from www.wireshark.org. Choose the software version you need based on your PC's architecture and operating system.

b. Follow the on-screen instructions to install Wireshark. If you are prompted to install USBPcap, **do NOT** install USBPcap for normal traffic capture. USBPcap is experimental, and it could cause USB problems on your PC.

Step 2: Capture DNS traffic.

a. Start Wireshark. Select an active interface with traffic for packet capture.

b. Clear the DNS cache.

1) In Windows, enter **ipconfig /flushdns** in Command Prompt.

2) For the majority of Linux distributions, one of the following utilities is used for DNS caching: Systemd - Resolved, DNSMasq, and NSCD. If your Linux distribution does not use one of the listed utilities, please perform an internet search for the DNS caching utility for your Linux distribution.

(i) Identify the utility used in your Linux distribution by checking the status:

Systemd-Resolved: **systemctl status systemd-resolved.service**

DNSMasq: **systemctl status dnsmasq.service**

NSCD: **systemctl status nscd.service**

(ii) If you are using system-resolved, enter **systemd-resolve --flush-caches** to flush the cache for Systemd-Resolved before restarting the service. The following commands restart the associated service using elevated privileges:

Systemd-Resolved: **sudo systemctl restart systemd-resolved.service**

DNSMasq: **sudo systemctl restart dnsmasq.service**

NSCD: **sudo systemctl restart nscd.service**

3) For the macOS, enter **sudo killall -HUP mDNSResponder** to clear the DNS cache in the Terminal. Perform an internet search for the commands to clear the DNS cache for an older OS.

c. At a command prompt or terminal, type **nslookup** enter the interactive mode.

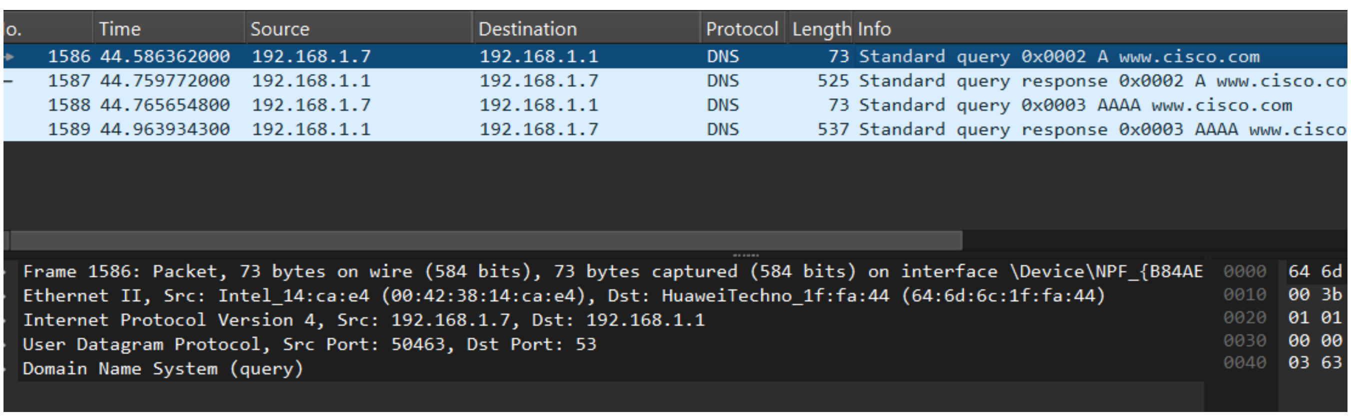
d. Enter the domain name of a website. The domain name www.cisco.com is used in this example.

e. Type **exit** when finished. Close the command prompt.

f. Click **Stop capturing packets** to stop the Wireshark capture.

Part 2: Explore DNS Query Traffic

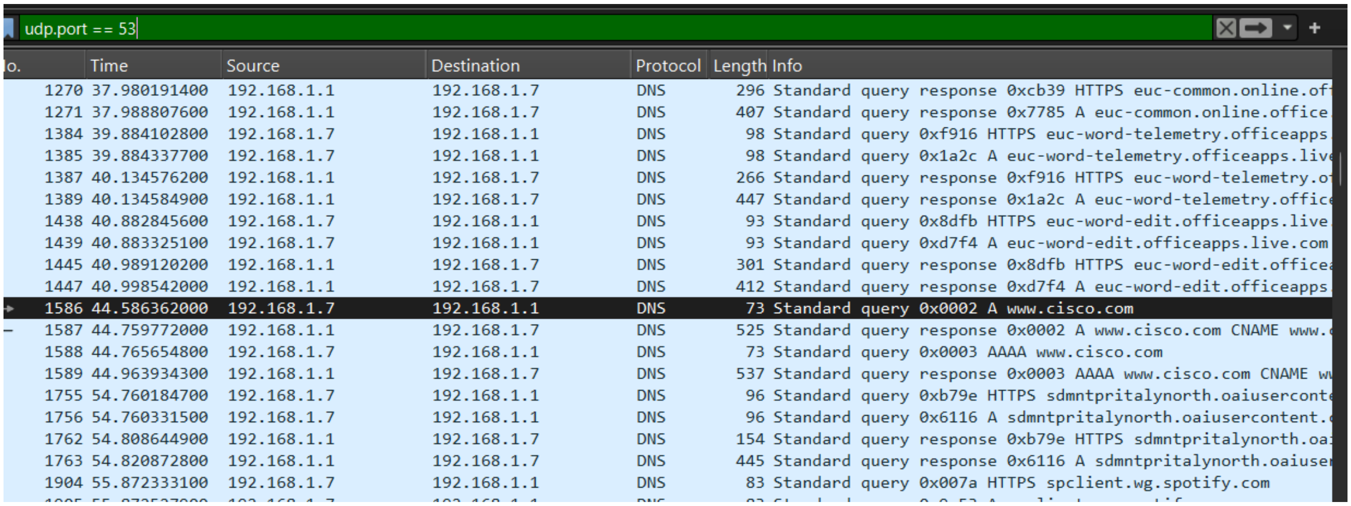
a. Observe the traffic captured in the Wireshark Packet List pane. Enter **udp.port == 53** in the filter box and click the arrow (or press enter) to display only DNS packets. **Note**: The provided screenshots are just examples. Your output maybe slightly different.



b. Select the DNS packet contains **Standard query** and **A www.cisco.com** in the Info column.

c. In the Packet Details pane, notice this packet has Ethernet II, Internet Protocol Version 4, User Datagram Protocol and Domain Name System (query).

d. Expand **Ethernet II** to view the details. Observe the source and destination fields.



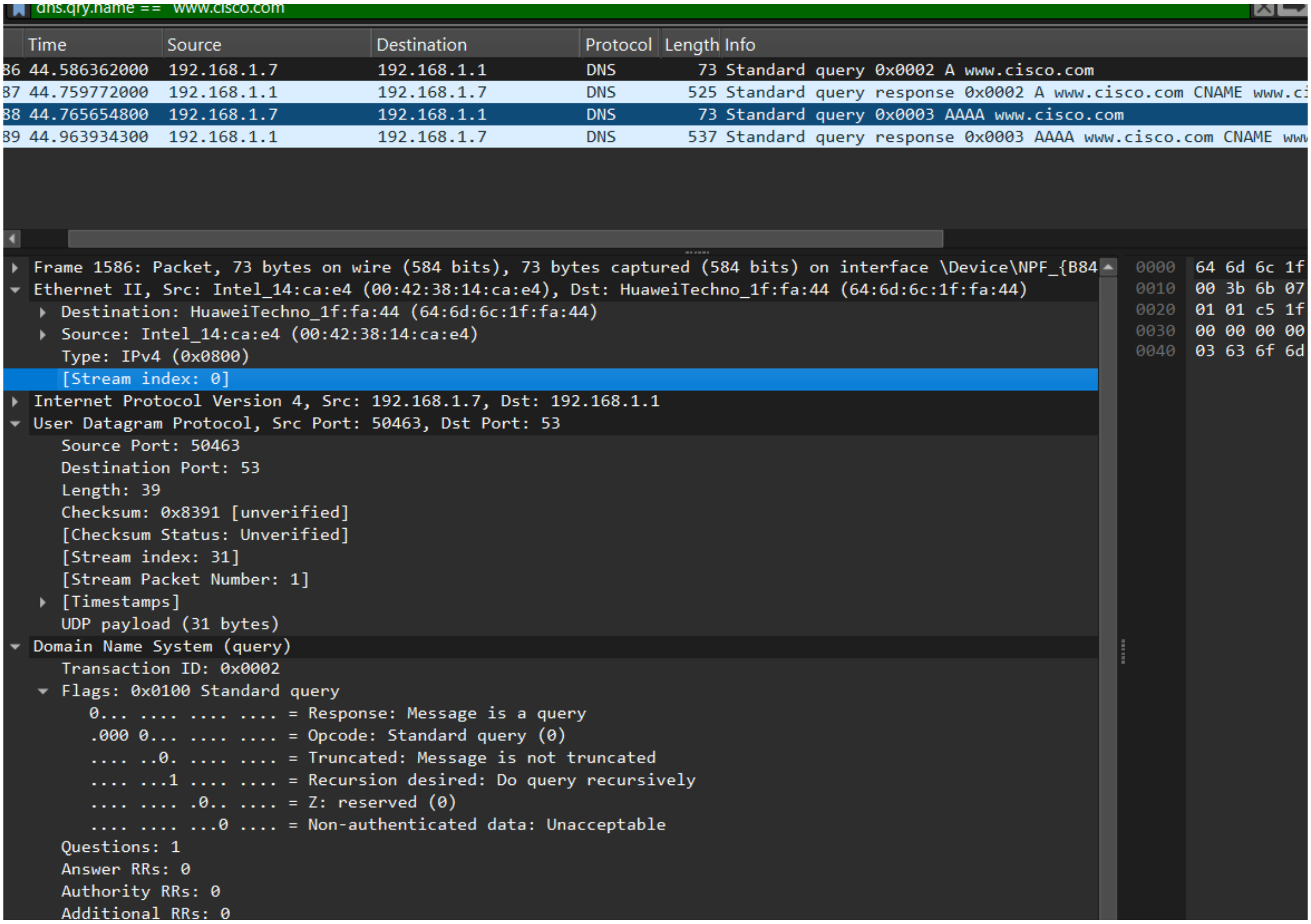
udp.port == 53

o.	Time	Source	Destination	Protocol	Length	Info
1270	37.980191400	192.168.1.1	192.168.1.7	DNS	296	Standard query response 0xcb39 HTTPS euc-common.online.of
1271	37.988807600	192.168.1.1	192.168.1.7	DNS	407	Standard query response 0x7785 A euc-common.online.office
1384	39.884102800	192.168.1.7	192.168.1.1	DNS	98	Standard query 0xf916 HTTPS euc-word-telemetry.officeapps
1385	39.884337700	192.168.1.7	192.168.1.1	DNS	98	Standard query 0x1a2c A euc-word-telemetry.officeapps.live
1387	40.134576200	192.168.1.1	192.168.1.7	DNS	266	Standard query response 0xf916 HTTPS euc-word-telemetry.o
1389	40.134584900	192.168.1.1	192.168.1.7	DNS	447	Standard query response 0x1a2c A euc-word-telemetry.offic
1438	40.882845600	192.168.1.7	192.168.1.1	DNS	93	Standard query 0x8dfb HTTPS euc-word-edit.officeapps.live
1439	40.883325100	192.168.1.7	192.168.1.1	DNS	93	Standard query 0xd7f4 A euc-word-edit.officeapps.live.com
1445	40.989120200	192.168.1.1	192.168.1.7	DNS	301	Standard query response 0x8dfb HTTPS euc-word-edit.office
1447	40.998542000	192.168.1.1	192.168.1.7	DNS	412	Standard query response 0xd7f4 A euc-word-edit.officeapps
1586	44.586362000	192.168.1.7	192.168.1.1	DNS	73	Standard query 0x0002 A www.cisco.com
1587	44.759772000	192.168.1.1	192.168.1.7	DNS	525	Standard query response 0x0002 A www.cisco.com CNAME www.
1588	44.765654800	192.168.1.7	192.168.1.1	DNS	73	Standard query 0x0003 AAAA www.cisco.com
1589	44.963934300	192.168.1.1	192.168.1.7	DNS	537	Standard query response 0x0003 AAAA www.cisco.com CNAME w
1755	54.760184700	192.168.1.7	192.168.1.1	DNS	96	Standard query 0xb79e HTTPS sdmntpritalynorth.oaiuserconte
1756	54.760331500	192.168.1.7	192.168.1.1	DNS	96	Standard query 0x6116 A sdmntpritalynorth.oaiusercontent.
1762	54.808644900	192.168.1.1	192.168.1.7	DNS	154	Standard query response 0xb79e HTTPS sdmntpritalynorth.oa
1763	54.820872800	192.168.1.1	192.168.1.7	DNS	445	Standard query response 0x6116 A sdmntpritalynorth.oaiuse
1904	55.872333100	192.168.1.7	192.168.1.1	DNS	83	Standard query 0x007a HTTPS spclient.wg.spotify.com

What are the source and destination MAC addresses? Which network interfaces are these MAC addresses associated with?

The source MAC address is `00:42:38:14:ca:e4`, associated with the PC's NIC. The destination MAC address is `64:6d:6c:1f:fa:44`, associated with the default gateway.

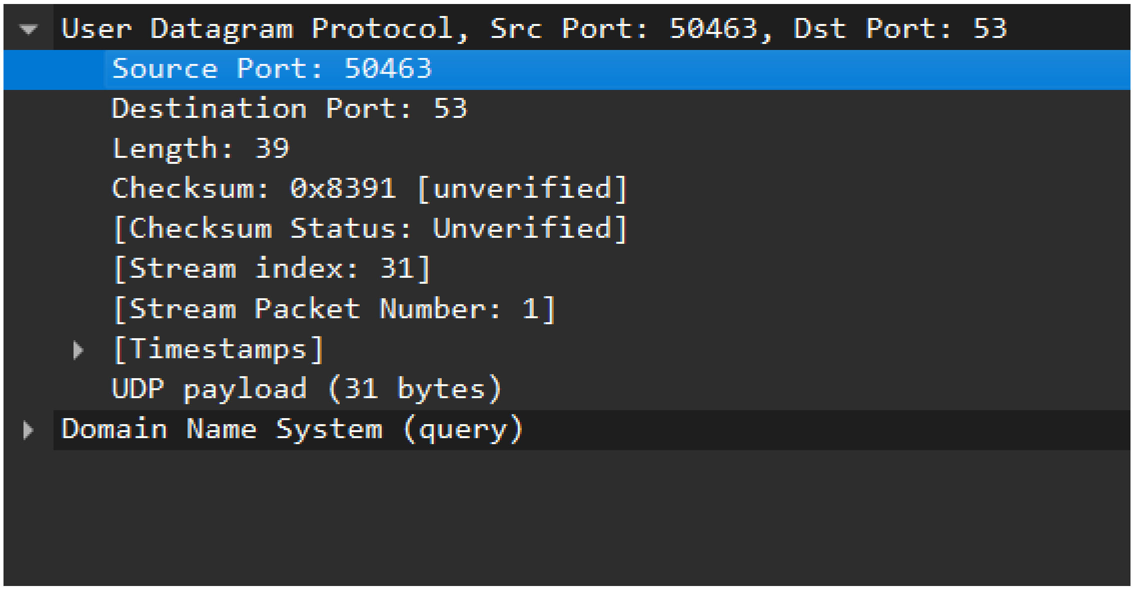
e. Expand **Internet Protocol Version 4**. Observe the source and destination IPv4 addresses.



What are the source and destination IP addresses? Which network interfaces are these IP addresses associated with?

The source IP address is 192.168.1.7, associated with the PC is network interface The destination IP address is 192.168.1.1, associated with the DNS server/default gateway

f. Expand the **User Datagram Protocol**. Observe the source and destination ports.



What are the source and destination ports? What is the default DNS port number?

- **e. The source port is `50463` and the destination port is `53`. The default DNS port number is `53`**

g. Determine the IP and MAC address of the PC.

1) In a Windows command prompt, enter **arp –a** and **ipconfig /all** to record the MAC and IP addresses of the PC.

2) For Linux and macOS PC, enter **ifconfig** or **ip address** in a terminal.

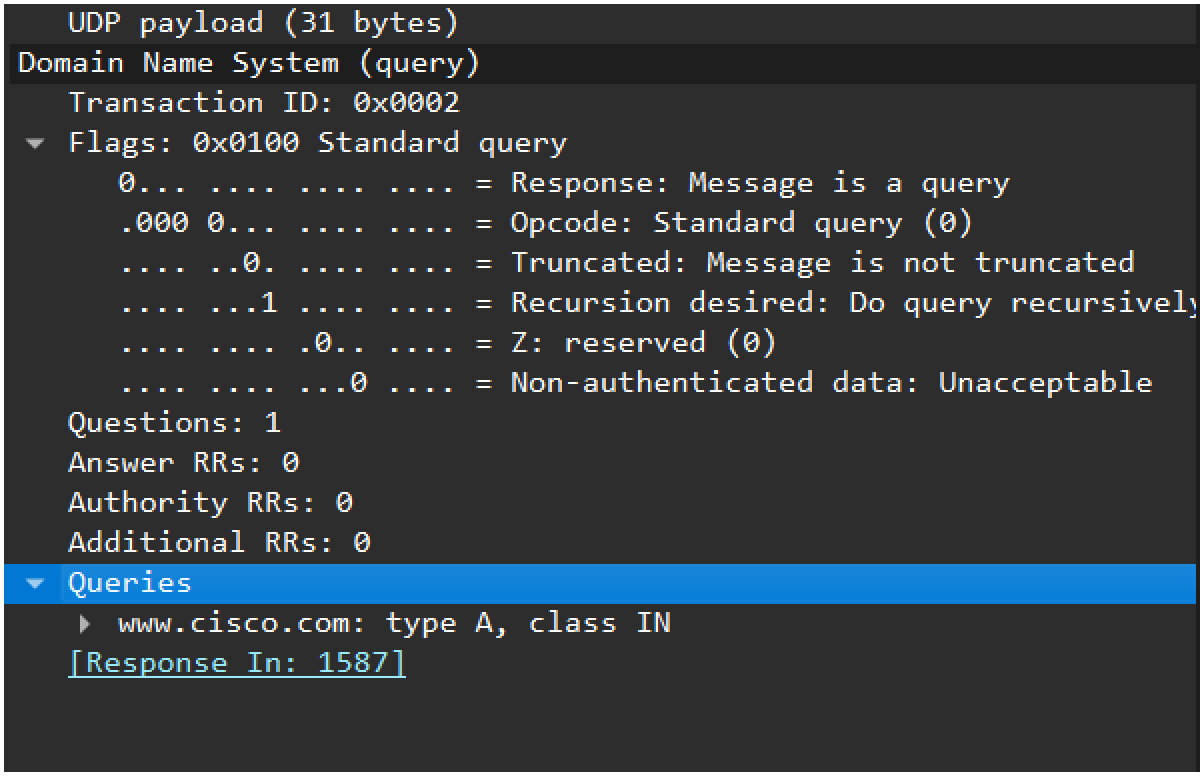
Compare the MAC and IP addresses in the Wireshark results to the IP and MAC addresses. What is your observation?

h. The IP and MAC addresses captured in the Wireshark results are the same

i. as the addresses listed in the `ipconfig /all` command.

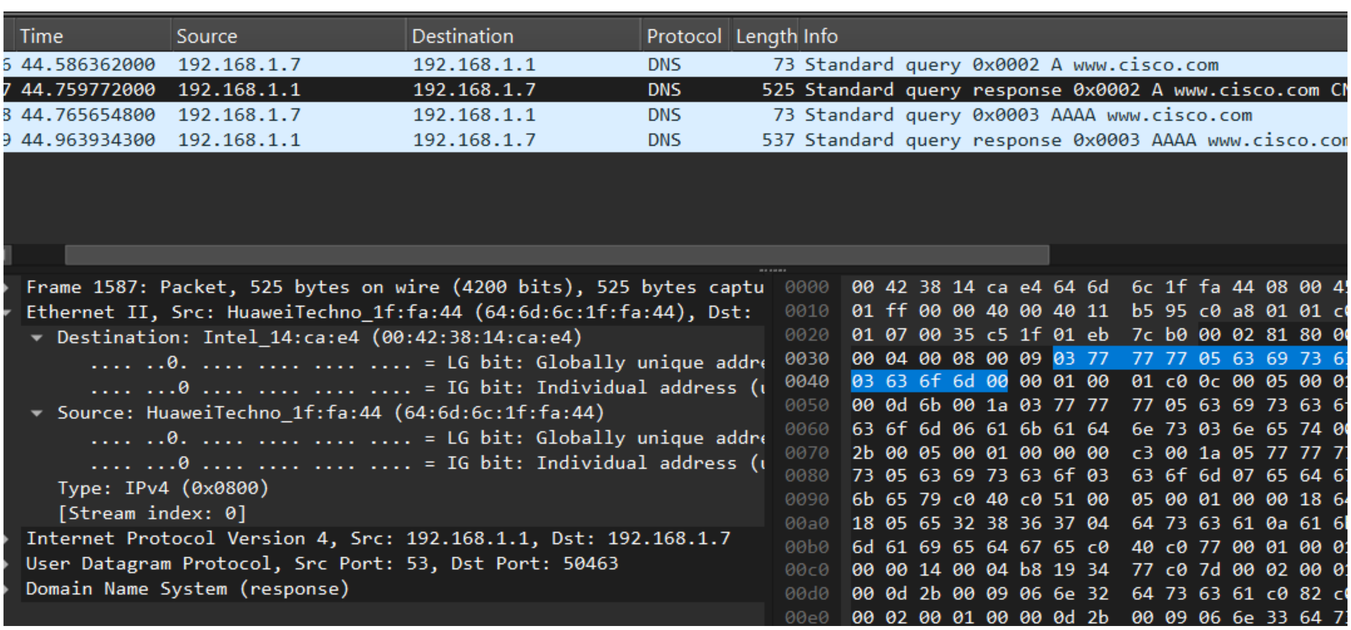
j. Expand **Domain Name System (query)** in the Packet Details pane. Then expand the **Flags** and **Queries**.

k. Observe the results. The flag is set to do the query recursively to query for the IP address to www.cisco.com.



Part 3: Explore DNS Response Traffic

a. Select the corresponding response DNS packet has **Standard query response** and **A www.cisco.com** in the Info column.



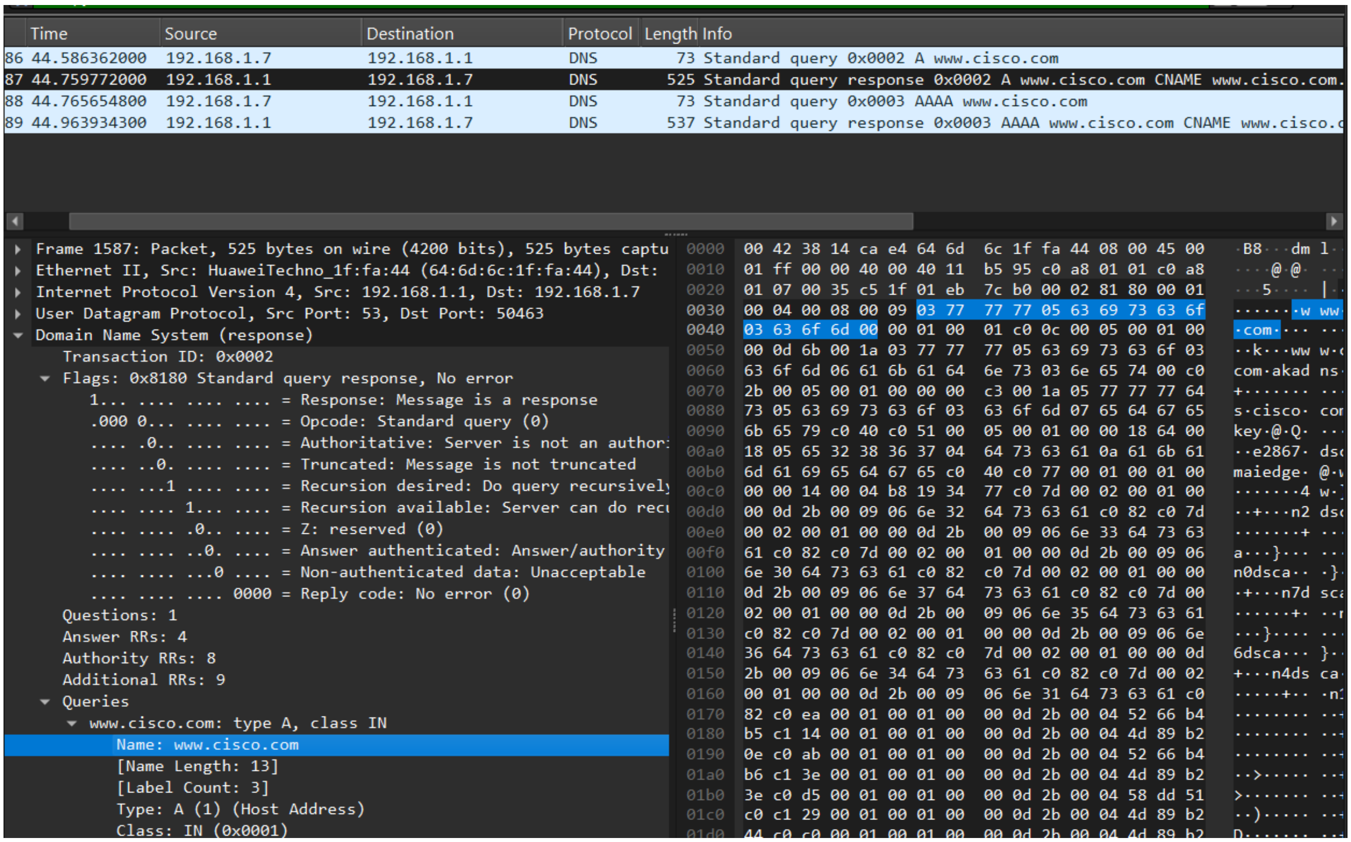
b. **Destination MAC:** `00:42:38:14:ca:e4`
 Source IP: `192.168.1.1`
 Destination IP: `192.168.1.7`
 Source Port: `53`
 Destination Port: `50463`

c. Expand **Domain Name System (response)**. Then expand the **Flags**, **Queries**, and **Answers**.

d. Observe the results.

Can the DNS server do recursive queries?

Yes, the DNS server can handle recursive queries.



e. Observe the CNAME and A records in the Answers details.

How do the results compare to nslookup results?

The results in Wireshark should be the same as the results from nslookup in the Command Prompt or terminal.

Reflection

1. From the Wireshark results, what else can you learn about the network when you remove the filter?

Without the filters, the results display other packets, such as DHCP and ARP. From these packets and the information contained within them, you can learn about other devices and their functions within the LAN.

2. How can an attacker use Wireshark to compromise your network security?

An attacker on the LAN can use Wireshark to observe the network traffic and can get sensitive information in the packet details if the traffic is not encrypted.